Uppala Prajin

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2026	8.36
NARAYANA JUNIOR COLLEGE, MEHDIPATNAM, HYDERABAD	Class XII (Telangana State Board of Intermediate Education)	2024	97.20%
DR KKR'S GOWTHAM CONCEPT HS ECIL X ROAD	Class X (Board of Secondary Education, Telangana)	2022	9.80%

PROJECTS

xv6 Kernel Extensions: Scheduling, Virtual Memory and Storage Systems (Coursework):

- Extended the xv6 operating system through a series of kernel-level assignments covering **system calls, scheduling, virtual memory, and storage systems**.
- Designed a virtual memory subsystem with lazy allocation, page fault handling, swapping, and Clock-based page replacement with priority-aware eviction.
- Implemented a **System Call Aware Multi-Level Feedback Queue (SC-MLFQ) scheduler** to distinguish **CPU-bound** vs **syscall-intensive** processes.
- Implemented disk-backed swapping using virtio disk with **RAID-0, RAID-1** and **RAID-5** support along with **FCFS** and **SSTF disk scheduling** policies storage simulation.
- Developed kernel-level system calls for process management, scheduling statistics, and memory monitoring with proper synchronization and locking.

Exploiting Floating Point Units for Integer Workloads:

- Optimized integer-heavy programs by leveraging **ALU-FPU parallelism** to increase **instruction-level parallelism** and speed up execution.
- Rewrote benchmarks including dot product, matrix multiplication, factorial, power, and prime checking using **loop unrolling** and **optimized integer-to-float ratios.**
- Simulated and validated performance improvements using Python scripts, analyzing execution trends across workloads.
- Evaluated performance on Apple M3 and Raspberry Pi, identifying workloads benefiting from **parallelism** versus **memory-bound** limitations.

Convex Clustering and Mixture Modeling using Sum-of-Norms Optimization:

- Studied convex clustering as a globally optimal alternative to K-means and GMMs by using KKT conditions.
- Extended the Sum-of-Norms framework with multiplicative weighting, comparing L1/L2 fusion for cluster recovery.
- Developed an ADMM-based Python implementation to validate agglomeration, anisotropic cluster recovery, and adaptive weighting strategies through simulations.
- Derived deterministic recovery guarantees using covariance-based bounds, reducing reliance on probabilistic concentration arguments and verified the results experimentally.

Arbitrary Precision Arithmetic Library in Java:

- Built an arbitrary-precision arithmetic library using **string-based representations** supporting integers and floating-point numbers, without relying on native fixed-width data types.
- Implemented addition, subtraction, multiplication, and division with sign handling and normalization using **long-division** and **partial-products algorithms**.
- Enabled high-precision floating-point division (up to 30 decimal places) **without native data type overflow**.
- Developed a command-line interface (CLI) with automated build support using Apache, Ant, Python and Git.

SKILLS

- **Programming Languages**: C, C++, Python, Java, RISC-V Assembly, SQL

- **Web Technologies**: , HTML, CSS, JavaScript, React
- **Databases and Tools**: PostgreSQL, Git, GitHub, MongoDB, Jupyter Notebook, Google Colab, LaTeX

RELEVANT COURSES

- **Computer Science:** Data Structures, Algorithms, Operating Systems, Database Management Systems, Computer Architecture, Theory of Computation, Software Development Fundamentals, Discrete Mathematics, Intro to Programming
- **Mathematics:** Probability and Random Variables, Calculus, Differential Equations, Convex Optimization, Linear Algebra, Transform Techniques, Random Processes, Statistics

POSITIONS OF RESPONSIBILITY

- **Teaching Assistant** for the course **Probability and Random Variables(AI1110)** under prof Anjana AM**, IIT Hyderabad**
- **Quant domain** Coordinator, **Finance and Consulting Club (FCC), IIT Hyderabad**
- **UG Mentor, Sunshine** (Counseling Cell), **IIT Hyderabad**
- **Web Head, Milan** (General Championship), **IIT Hyderabad**
- **Web Coordinator, Milan** (General Championship), **IIT Hyderabad**

SCHOLASTIC ACHIEVEMENTS

- **Specialist** on **Codeforces** (max rating - **1443**)